

DATA HANDLING AND VISUALIZATION (DSA06)

EXPERIMENTS 1 TO 12 REPORT

Experiment 1: Monthly Sales Line Chart

Question / Task:

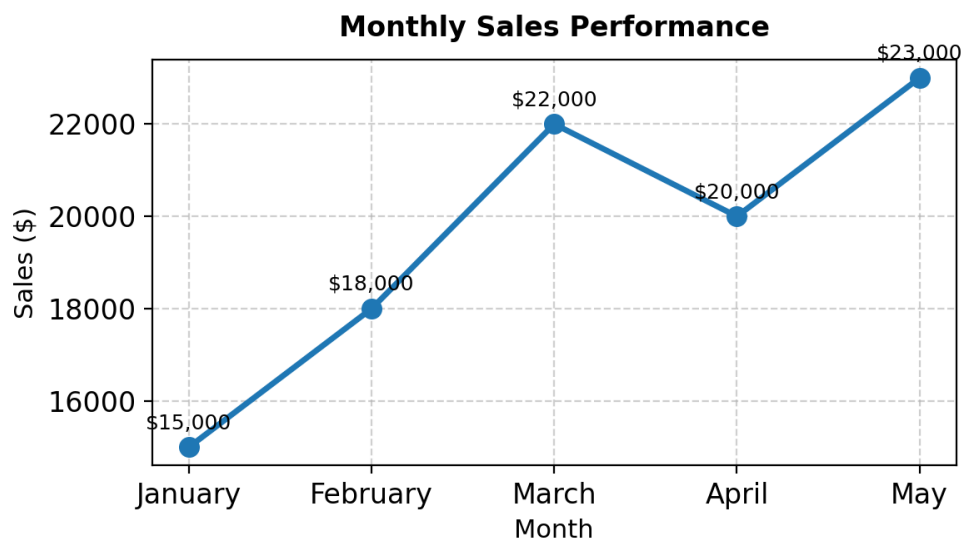
Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

R Program Code:

```
# Experiment 1: Monthly Sales Line Chart
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)
df <- data.frame(month = factor(month, levels = month), sales = sales)

plot(df$sales, type = "o", col = "#1F74B4", pch = 19, lwd = 2.5,
     main = "Monthly Sales Performance",
     xlab = "Month", ylab = "Sales (in $)", xaxt = "n")
axis(1, at = 1:5, labels = df$month)
grid(nx = NULL, ny = NULL, col = "lightgray", lty = "dotted")
text(1:5, df$sales + 400, labels = paste0("$", df$sales), cex = 0.8)
```

Output Screenshot:



Experiment 2: Top-Selling Products Bar Chart

Question / Task:

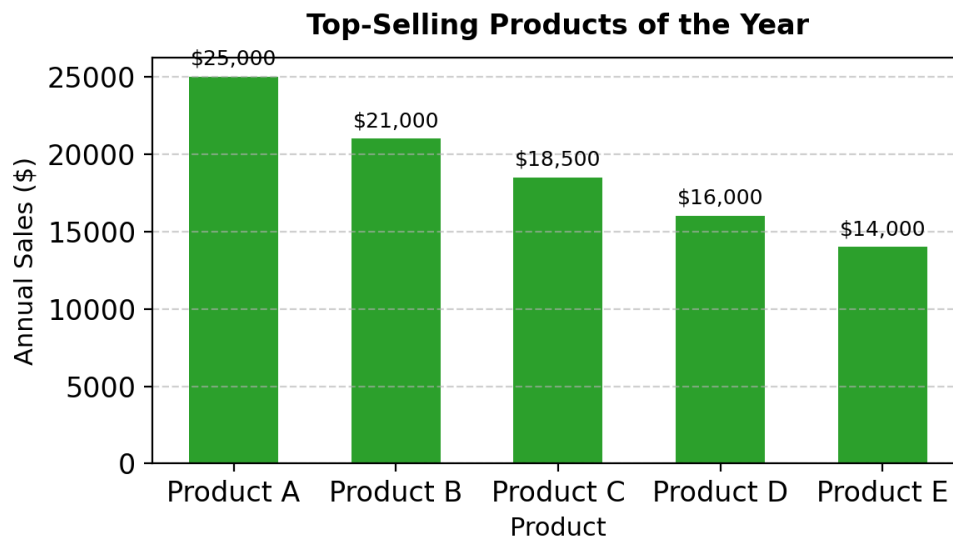
Generate a bar chart to display the top-selling products for the year. Label the chart elements.

R Program Code:

```
# Experiment 2: Top-Selling Products Bar Chart
products <- c("Product A", "Product B", "Product C", "Product D", "Product E")
sales <- c(25000, 21000, 18500, 16000, 14000)

bp <- barplot(sales, names.arg = products, col = "#2CA02C",
              main = "Top-Selling Products of the Year",
              xlab = "Product", ylab = "Annual Sales ($)",
              ylim = c(0, 30000))
text(bp, sales + 1000, labels = paste0("$", sales), cex = 0.8)
grid(nx = NA, ny = NULL, col = "lightgray", lty = "dotted")
```

Output Screenshot:



Experiment 3: Advertising Budget vs Monthly Sales Scatter Plot

Question / Task:

Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.

R Program Code:

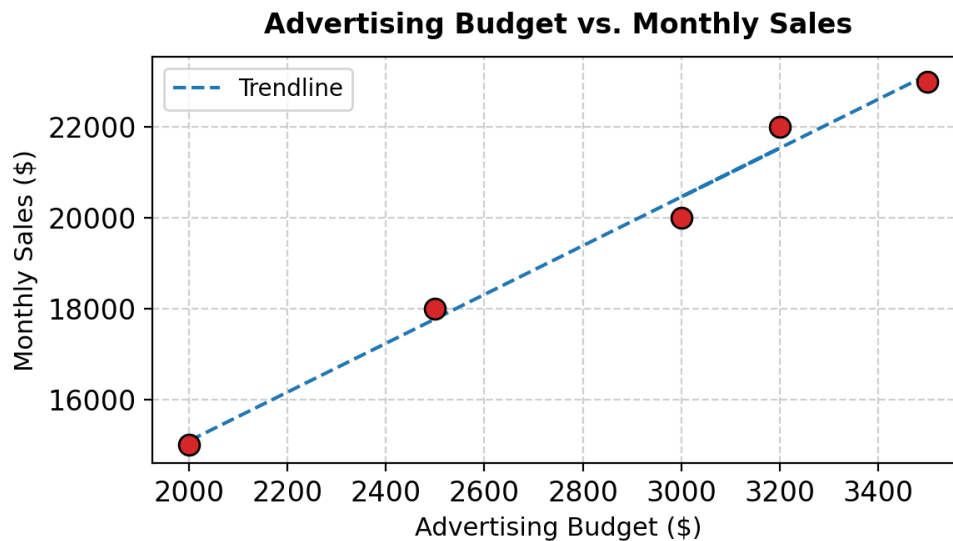
```
# Experiment 3: Scatter Plot & Trend Analysis
budget <- c(2000, 2500, 3200, 3000, 3500)
sales <- c(15000, 18000, 22000, 20000, 23000)

plot(budget, sales, col = "#D62728", pch = 19, cex = 1.5,
     main = "Advertising Budget vs. Monthly Sales",
     xlab = "Advertising Budget ($)", ylab = "Monthly Sales ($)")
abline(lm(sales ~ budget), col = "#1F74B4", lwd = 2, lty = 2)
grid(nx = NULL, ny = NULL, col = "lightgray", lty = "dotted")

# Insight Text Output
```

```
cat("Insight: A strong positive linear correlation exists between advertising budget and monthly sales.")
```

Output Screenshot:



Experiment 4: Interactive Sales Dashboard Layout

Question / Task:

Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

R Program Code:

```
# Experiment 4: Interactive Dashboard Layout
par(mfrow = c(1, 2))

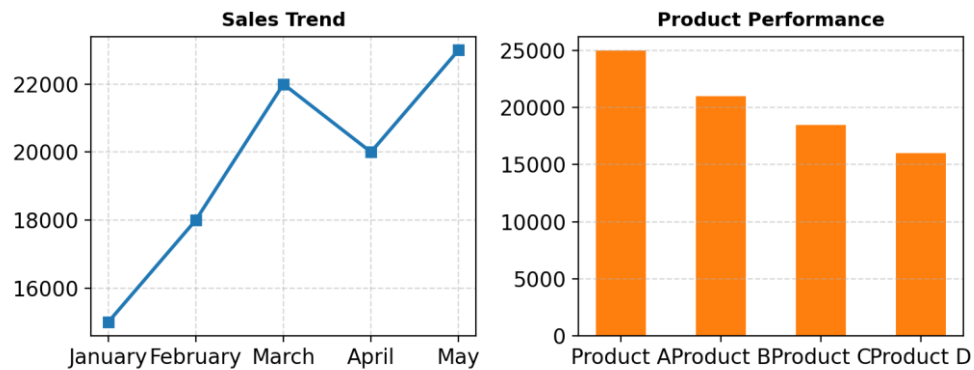
# Chart 1: Line Plot (Sales Trend)
month <- c("Jan", "Feb", "Mar", "Apr", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)
plot(sales, type = "o", col = "#1F74B4", pch = 15, main = "Sales Trend",
     xlab = "Month", ylab = "Sales ($)", xaxt = "n")
axis(1, at = 1:5, labels = month)

# Chart 2: Bar Plot (Product Sales)
products <- c("Prod A", "Prod B", "Prod C", "Prod D")
p_sales <- c(25000, 21000, 18500, 16000)
barplot(p_sales, names.arg = products, col = "#FF7F0E", main = "Product Performance",
       xlab = "Product", ylab = "Sales ($)")

par(mfrow = c(1, 1))
```

Output Screenshot:

Interactive Sales Dashboard View



Experiment 5: Customer Age Distribution Histogram

Question / Task:

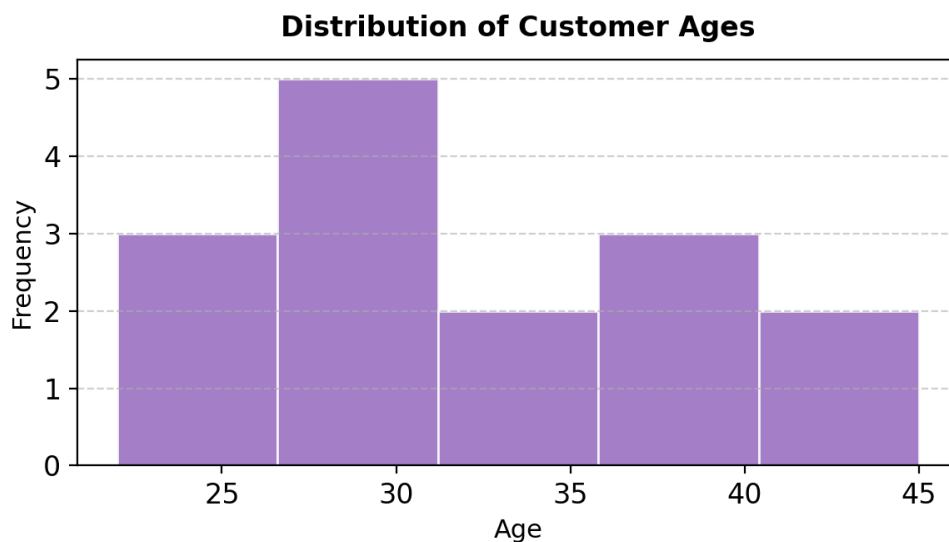
Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

R Program Code:

```
# Experiment 5: Customer Age Histogram
ages <- c(25, 30, 35, 28, 40, 22, 31, 29, 38, 45, 27, 33, 36, 41, 24)

hist(ages, col = "#9467BD", border = "white",
     main = "Distribution of Customer Ages",
     xlab = "Age", ylab = "Frequency", breaks = 5)
grid(nx = NA, ny = NULL, col = "lightgray", lty = "dotted")
```

Output Screenshot:



Experiment 6: Customer Satisfaction Score Pie Chart

Question / Task:

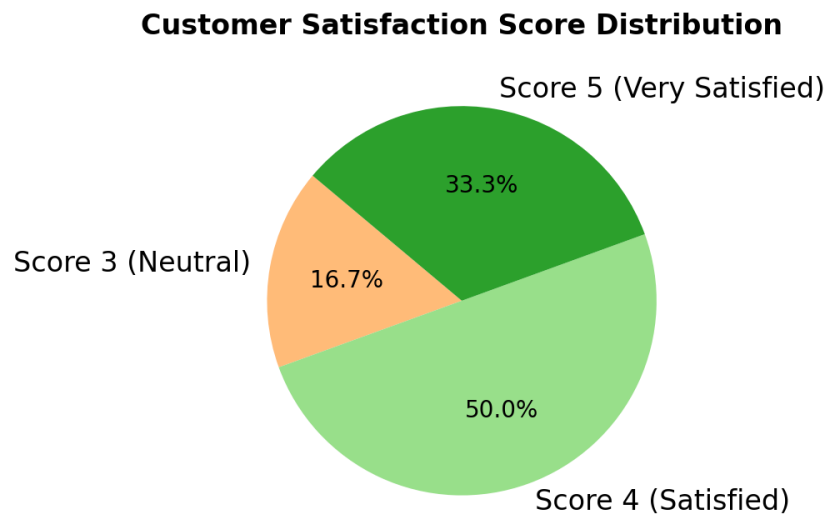
Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

R Program Code:

```
# Experiment 6: Customer Satisfaction Pie Chart
scores_count <- c(1, 3, 2)
labels <- c("Score 3 (Neutral)", "Score 4 (Satisfied)", "Score 5 (Very Satisfied)")
pct <- round(scores_count/sum(scores_count)*100, 1)
lbls <- paste(labels, pct, "%", sep=" ")

pie(scores_count, labels = lbls, col = c("#FFBB78", "#98DF8A", "#2CA02C"),
    main = "Customer Satisfaction Score Distribution")
```

Output Screenshot:



Experiment 7: Satisfaction Score by Age Group Stacked Bar Chart

Question / Task:

Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

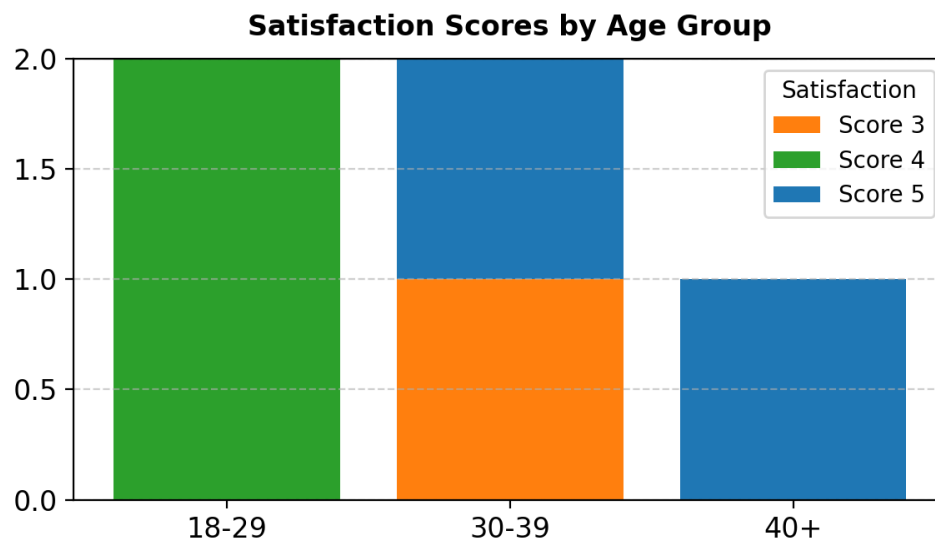
R Program Code:

```
# Experiment 7: Stacked Bar Chart
data_matrix <- matrix(c(0, 2, 0,
                        1, 0, 1,
                        0, 0, 1),
                      nrow = 3, byrow = FALSE)
colnames(data_matrix) <- c("18-29", "30-39", "40+")
rownames(data_matrix) <- c("Score 3", "Score 4", "Score 5")

barplot(data_matrix, col = c("#FF7F0E", "#2CA02C", "#1F74B4"),
    main = "Satisfaction Scores by Age Group",
```

```
xlab = "Age Group", ylab = "Count", legend.text = TRUE,
args.legend = list(x = "topright", bty = "n"))
```

Output Screenshot:



Experiment 8: Customer Feedback Word Cloud

Question / Task:

Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

R Program Code:

```
# Experiment 8: Feedback Sentiment Word Cloud
library(wordcloud)
words <- c("Excellent", "Good Service", "Fast Support", "User Friendly", "Reliable",
"Needs Improvement", "Great Interface", "Helpful")
freq <- c(50, 40, 35, 30, 28, 15, 25, 20)

set.seed(123)
wordcloud(words = words, freq = freq, min.freq = 1,
          scale = c(3, 0.5), colors = brewer.pal(8, "Dark2"),
          random.order = FALSE)
```

Output Screenshot:

Customer Feedback Word Cloud



Experiment 9: Employee Performance Trend Line Chart

Question / Task:

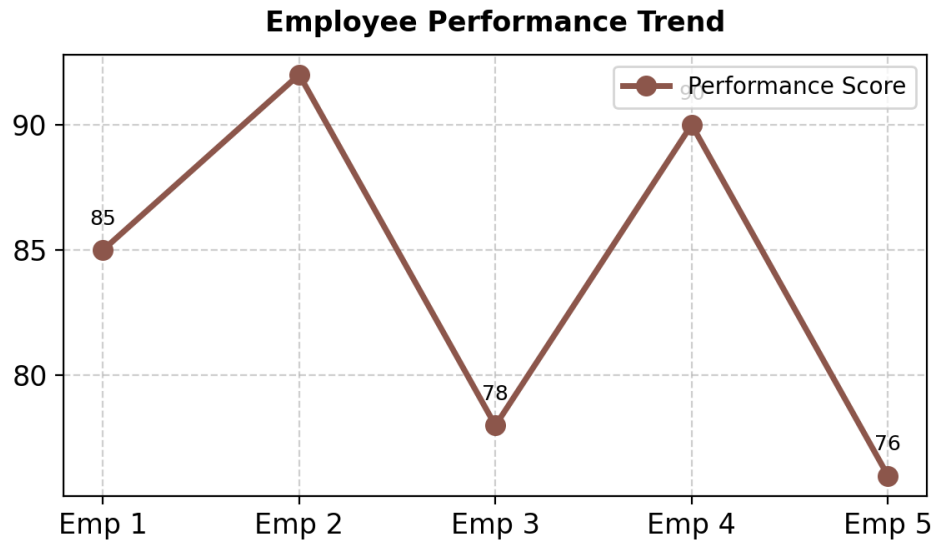
Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

R Program Code:

```
# Experiment 9: Employee Performance Line Chart
emp_id <- c("Emp 1", "Emp 2", "Emp 3", "Emp 4", "Emp 5")
scores <- c(85, 92, 78, 90, 76)

plot(scores, type = "o", col = "#8C564B", pch = 19, lwd = 2.5,
      main = "Employee Performance Trend",
      xlab = "Employee ID", ylab = "Performance Score", xaxt = "n", ylim = c(70, 100))
axis(1, at = 1:5, labels = emp_id)
grid(nx = NULL, ny = NULL, col = "lightgray", lty = "dotted")
legend("topright", legend = "Performance Score", col = "#8C564B", lty = 1, pch = 19)
text(1:5, scores + 2, labels = scores, cex = 0.8)
```

Output Screenshot:



Experiment 10: Department Distribution Bar Chart

Question / Task:

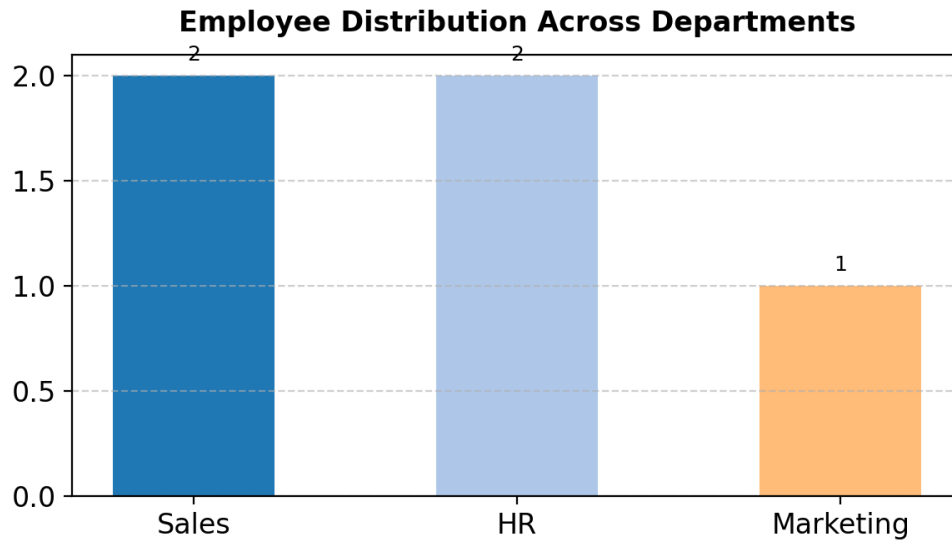
Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

R Program Code:

```
# Experiment 10: Department Distribution Bar Chart
dept_counts <- table(c("Sales", "HR", "Marketing", "Sales", "HR"))

bp <- barplot(dept_counts, col = c("#1F74B4", "#AEC7E8", "#FFBB78"),
              main = "Employee Distribution Across Departments",
              xlab = "Department", ylab = "Number of Employees",
              ylim = c(0, 3))
text(bp, dept_counts + 0.2, labels = dept_counts, cex = 0.9)
grid(nx = NA, ny = NULL, col = "lightgray", lty = "dotted")
```

Output Screenshot:



Experiment 11: Years of Service vs Performance Score Scatter Plot

Question / Task:

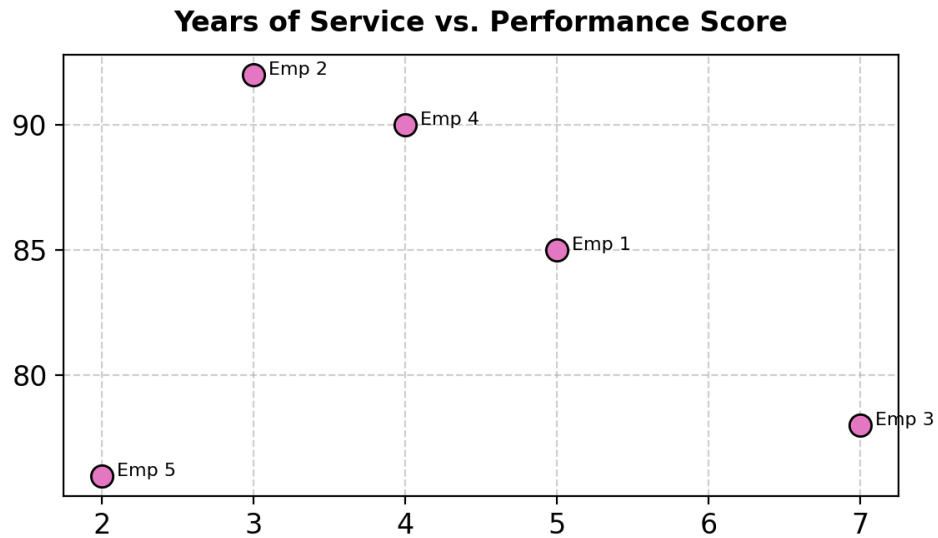
Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

R Program Code:

```
# Experiment 11: Service vs Performance Scatter Plot
service <- c(5, 3, 7, 4, 2)
scores <- c(85, 92, 78, 90, 76)

plot(service, scores, col = "#E377C2", pch = 19, cex = 1.8,
      main = "Years of Service vs. Performance Score",
      xlab = "Years of Service", ylab = "Performance Score")
text(service + 0.15, scores, labels = paste("Emp", 1:5), cex = 0.8, pos = 4)
grid(nx = NULL, ny = NULL, col = "lightgray", lty = "dotted")
```

Output Screenshot:



Experiment 12: Interactive Tableau Dashboard View

Question / Task:

Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

R Program Code:

```
# Experiment 12: Dashboard View Simulation
par(mfrow = c(1, 2))

# Subplot 1: Trend
emp_id <- c("Emp 1", "Emp 2", "Emp 3", "Emp 4", "Emp 5")
scores <- c(85, 92, 78, 90, 76)
plot(scores, type = "o", col = "#8C564B", pch = 19, main = "Performance Trend",
      xlab = "Employee ID", ylab = "Score", xaxt = "n")
axis(1, at = 1:5, labels = emp_id)

# Subplot 2: Department Count
dept_counts <- c(Sales = 2, HR = 2, Marketing = 1)
barplot(dept_counts, col = "#1F74B4", main = "Department Headcount",
        xlab = "Department", ylab = "Count")

par(mfrow = c(1, 1))
```

Output Screenshot:

Tableau Dashboard: Employee Performance Analysis

